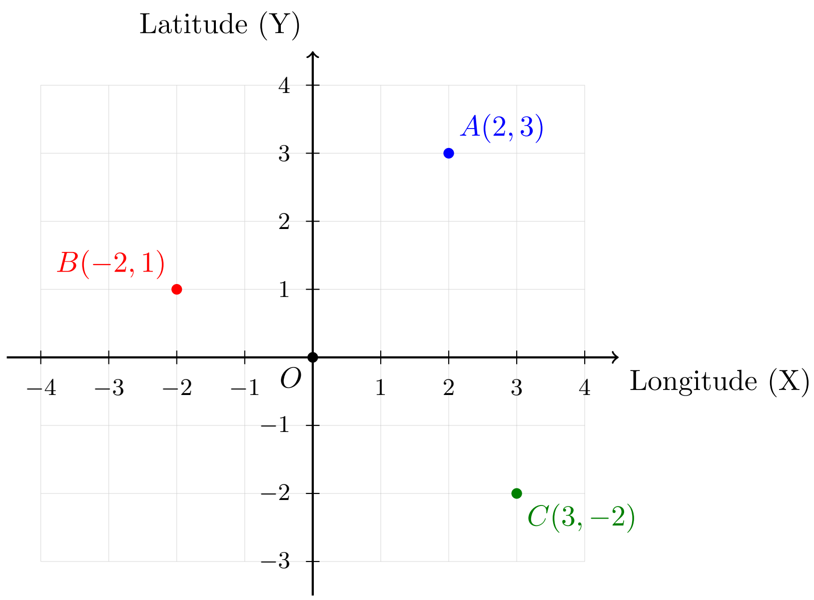
Practice Sheet & Notes:
2D Vectors, Dot-product and Cosine Similarity

Study Notes

1. Specifying 2D Points (Latitude and Longitude)

In a 2D Cartesian coordinate system, a point is specified by a pair of values (x, y). For geographical or spatial contexts, the X-axis typically represents **Longitude** (East-West position), and the Y-axis represents **Latitude** (North-South position). The point **O** represents the Origin, or $(0, 0)$.

Example Graph: Let us plot three points: $A(2, 3)$, $B(-2, 1)$, and $C(3, -2)$.



2. Euclidean Distance Between Two Points

The Euclidean distance d between two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is given by the formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example Calculation: Find the distance between $A(2, 3)$ and $B(-2, 1)$.

$$d(A, B) = \sqrt{(-2 - 2)^2 + (1 - 3)^2} = \sqrt{(-4)^2 + (-2)^2} = \sqrt{16 + 4} = \sqrt{20} = 2\sqrt{5} \approx 4.47$$

3. Norm (Magnitude) of Position Vectors

When a point $P(x, y)$ is treated as a position vector $\vec{P}$ starting from the origin $(0, 0)$, its **norm** (or length) is denoted by $\|\vec{P}\|$.

$$\|\vec{P}\| = \sqrt{x^2 + y^2}$$

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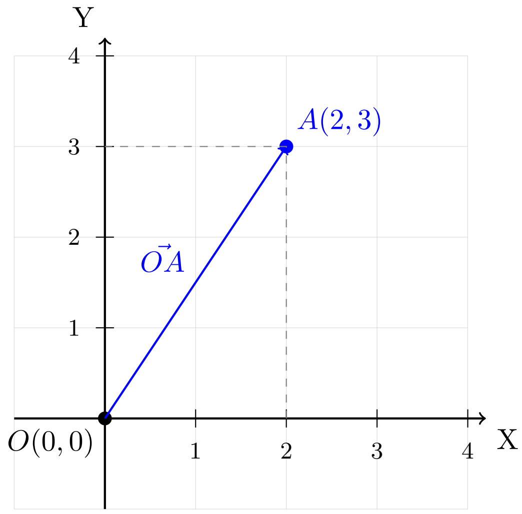
Example Calculation: Find the norm of the position vector for $A(2,3)$.

$$\|\vec{A}\| = \sqrt{2^2 + 3^2} = \sqrt{4+9} = \sqrt{13} \approx 3.61$$



The length of the point vector $\vec{OA}$ is ≈ 3.61

4. Dot Product of Position Vectors

The dot product of two position vectors $\vec{U} = \langle x_1, y_1 \rangle$ and $\vec{V} = \langle x_2, y_2 \rangle$ is a scalar value calculated as:

$$\vec{U} \cdot \vec{V} = (x_1 \times x_2) + (y_1 \times y_2)$$

Example Calculation: Find the dot product of $\vec{A} = \langle 2, 3 \rangle$ and $\vec{B} = \langle -2, 1 \rangle$.

$$\vec{A} \cdot \vec{B} = (2 \times -2) + (3 \times 1) = -4 + 3 = -1$$

5. Cosine Similarity Score

The cosine similarity score measures how aligned two vectors are, independent of their magnitude. It represents the cosine of the angle θ between the points.

$$\text{Cosine Similarity} = \cos(\theta) = \frac{\vec{U} \cdot \vec{V}}{\|\vec{U}\|\|\vec{V}\|}$$

Note: The similarity values have a cap (maximum) value of 1. A value of 1 indicates a perfectly overlapping orientation (the vectors point in the exact same direction). A value of 0 means the vectors are wide apart at exactly 90° (orthogonal). Negative values indicate they point in opposing directions.

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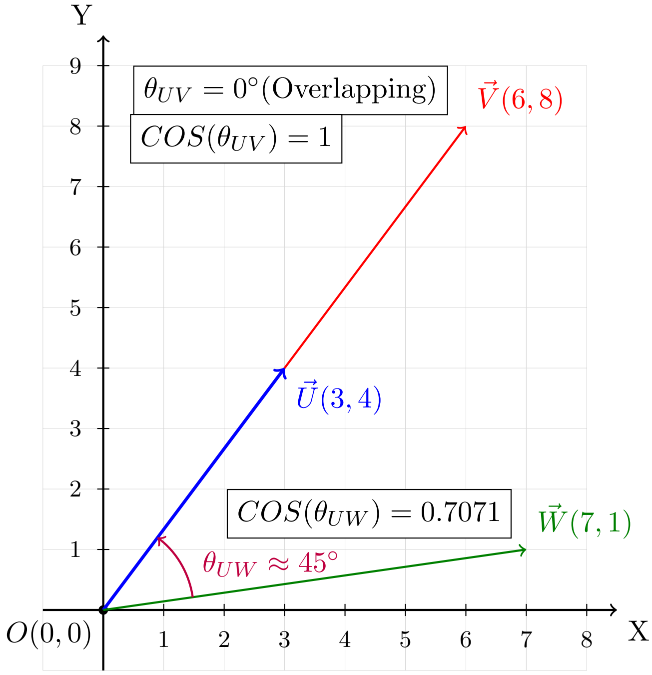

Example Calculation 1 (Overlapping vectors):
Let $\vec{U} = \langle 3, 4 \rangle$ and $\vec{V} = \langle 6, 8 \rangle$.

$$\vec{U} \cdot \vec{V} = (3)(6) + (4)(8) = 18 + 32 = 50$$

$$\|\vec{U}\| = \sqrt{3^2 + 4^2} = 5, \quad \|\vec{V}\| = \sqrt{6^2 + 8^2} = 10$$

$$\text{COSINE}(\vec{U}, \vec{V}) = \frac{50}{5 \times 10} = \frac{50}{50} = 1$$

Result: Cosine score = 1 (Perfectly overlapping orientation; or highly **similar**).

Example Calculation 2 (dissimilar vectors):
Let $\vec{U} = \langle 3, 4 \rangle$ and $\vec{W} = \langle 7, 1 \rangle$.

$$\vec{U} \cdot \vec{W} = (3)(7) + (4)(1) = 21 + 4 = 25$$

$$\|\vec{U}\| = \sqrt{3^2 + 4^2} = 5, \quad \|\vec{W}\| = \sqrt{7^2 + 1^2} = \sqrt{50} \approx 7.1$$

$$\text{COSINE}(\vec{U}, \vec{W}) = \frac{25}{5 \times 7.1} = \frac{50}{35.5} = 0.7071$$

Result: Cosine score = 0.7071 (**slightly dissimilar**).

Example Calculation 3 (Orthogonal vectors):
Let $\vec{X} = \langle 1, 0 \rangle$ and $\vec{Y} = \langle 0, 1 \rangle$.

$$\vec{X} \cdot \vec{Y} = (1)(0) + (0)(1) = 0$$

$$\text{COSINE}(\vec{X}, \vec{Y}) = \frac{0}{1 \times 1} = 0$$

Result: Cosine score = 0 (Wide apart, 90° separation; or highly **dissimilar**).

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Practice Questions

Q1: Specifying Coordinates
Let three locations on a map be given by their (Longitude, Latitude) coordinates: $P(3,4)$, $Q(-1,2)$, and $R(4,-3)$.

(a) Draw a Cartesian plane, label the axes appropriately (Longitude / Latitude), and plot points P, Q, and R.

(b) Express P, Q, and R as position vectors originating from $(0,0)$.

Q2: Euclidean Distance
Using the points from Q1:

(a) Calculate the straight-line Euclidean distance between P and Q.

(b) Calculate the straight-line Euclidean distance between Q and R.

Q3: Norm Computations
Calculate the norm (magnitude) of the following position vectors:

(a) $\|\vec{P}\|$

(b) $\|\vec{Q}\|$

(c) $\|\vec{R}\|$

Q4: Dot Products
Compute the dot product for the following pairs of position vectors:

(a) $\vec{P} \cdot \vec{Q}$

(b) $\vec{Q} \cdot \vec{R}$

(c) $\vec{P} \cdot \vec{R}$

Q5: Cosine Similarity
Calculate the cosine similarity score between the following pairs. (Leave answers in exact fraction/radical form if necessary).

(a) $\vec{P}$ and $\vec{Q}$

(b) $\vec{Q}$ and $\vec{R}$

Q6: Orthogonality (90-Degree Separation)
A new point S has coordinates $(x,6)$.

(a) Write the position vector for S in terms of x.

(b) Find the value of x such that $\vec{S}$ is exactly $90°$ apart from $\vec{P}(3,4)$ (i.e., their cosine similarity score is 0).

Q7: Overlapping Orientations
Consider a point $T(9,12)$.

(a) Calculate the cosine similarity score between $\vec{P}(3,4)$ and $\vec{T}$.

(b) Interpret the physical meaning of this score in terms of their orientation.

Q8: Geometric Application
Consider the vectors $\vec{M} = \langle 5,2\rangle$ and $\vec{N} = \langle -2,5\rangle$.

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(a) Calculate their dot product and the cosine similarity score.

(b) Based on the cosine similarity score, what is the exact angle between $\vec{M}$ and $\vec{N}$?

(c) Find the Euclidean distance between points M and N.

(d) Verify that the triangle formed by the Origin $(0, 0)$, M, and N satisfies the Pythagorean theorem ($OM^2 + ON^2 = MN^2$).